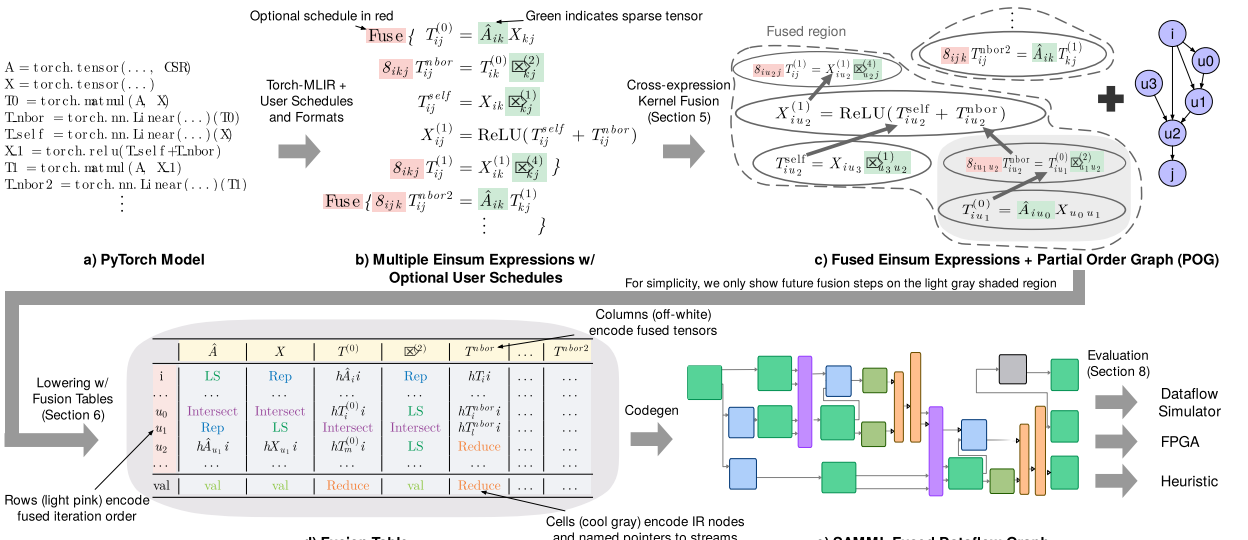

A = torch.tensor(..., CSR)
X = torch.tensor(...)
T0 = torch.matmul(A, X)
Tnbor = torch.nn.Linear(...)(T0)
Tself = torch.nn.Linear(...)(X)
X1 = torch.relu(Tself+Tnbor)
T1 = torch.matmul(A, X1)
Tnbor2 = torch.nn.Linear(...)(T1)
Torch-MLIR + User Schedules and Formats
Optional schedule in red
Green indicates sparse tensor
Fuse
Cross-expression Kernel Fusion (Section 5)
Fused region
a) PyTorch Model
b) Multiple Einsum Expressions w/ Optional User Schedules
c) Fused Einsum Expressions + Partial Order Graph (POG)
For simplicity, we only show future fusion steps on the light gray shaded region
Lowering w/ Fusion Tables (Section 6)
Columns (off-white) encode fused tensors
Rows (light pink) encode fused iteration order
Cells (cool gray) encode IR nodes and named pointers to streams
Codegen
Evaluation (Section 8)
Dataflow Simulator
FPGA
Heuristic
d) Fusion Table
e) SAMML Fused Dataflow Graph